Rudra Prasadh Ganesh

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SUMMARY

Software engineer with production experience across robotics, full-stack development, AI/ML systems, cloud infrastructure, and embedded systems. Built and shipped real-time platforms, autonomous robot software, analytics dashboards, data pipelines, and scalable APIs in startup, enterprise, and research environments. Peer-reviewed publication at AIED 2026 (Springer). Lee Kuan Yew Gold Medal recipient, top graduating student, CGPA 4.94 / 5.00, Honours (Highest Distinction).

EDUCATION

Nanyang Technological University, Singapore
Bachelor of Engineering, Computer Engineering | Singapore | Aug 2022 - May 2026

- CGPA: 4.94 / 5.00
- Degree Classification: Honours (Highest Distinction)
- Elective Focus Awarded: Artificial Intelligence
- Lee Kuan Yew Gold Medal - awarded to the top graduating student of the cohort.
- NTU Science and Engineering Undergraduate Scholar - Full Scholarship.
- Dean's List in all four academic years. Top 5 percent of cohort.

Sri KDU International School, Malaysia
IB Diploma, 100 percent Tuition Scholar | Malaysia | Sep 2020 - May 2022

- Score: 42 / 45

Global Indian International School, Kuala Lumpur
CBSE Class 10 | Malaysia | Apr 2019 - Mar 2020

- Score: 96.2 percent

SKILLS

Programming Languages: Python, TypeScript, JavaScript, Java, C#, SQL, C, C++, Bash, HTML, CSS, PHP, Solidity, ARM Assembly

Frontend Engineering: React.js, Next.js, React Native, Expo, Redux, Tailwind CSS, Vite, Webpack, responsive UI, state management, component architecture

Backend Engineering: FastAPI, Django, Django REST Framework, Node.js, Express.js, ASP.NET, Spring Boot, REST API design, WebSockets, authentication, authorization, JWT, OAuth

Databases and Caching: PostgreSQL, MySQL, MongoDB, Firebase, Firestore, Redis, SQL query optimization, schema design, indexing

Cloud and DevOps: Alibaba Cloud, AWS, AWS Lambda, Amazon S3, AWS Elastic Beanstalk, Amazon SageMaker, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Vercel, deployment automation

Data and Analytics: ETL, ELT, data modeling, feature engineering, Pandas, NumPy, PostGIS, Power BI, Tableau, dashboard design, time-series forecasting

AI/ML and LLM Systems: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, Transformers, LoRA, PEFT, RAG, LangChain, LangGraph, Ollama, prompt engineering, fine-tuning, model evaluation

Computer Vision and NLP: YOLO, YOLOv8, Grad-CAM, tokenization, sentiment analysis, multimodal integration, ESM-2, AlphaFold

Robotics: ROS, ROS2, Robot Operating System, robot perception, robot control, autonomous navigation, simulation, reinforcement learning, LIDAR, computer vision, sensor fusion

Embedded and Hardware: FreeRTOS, STM32, STM32F4, XMC4800, MSP432, ESP32, Arduino, ODROID, EtherCAT, CAN bus, CAN, I2C, SPI, UART, sensor integration, motor control, real-time scheduling, firmware debugging

Blockchain and Web3: Solidity, Hardhat, Ethers.js, Web3.js, Ethereum, XRPL, IPFS, smart contract integration

Software Practices: system design, debugging, performance optimization, testing, observability, reliability engineering, agile, Scrum, code review

Tools: Git, GitHub, Linux, JIRA, Postman, VS Code, Figma

EXPERIENCE

Fabrica AI

Software Engineer | Singapore | Jul 2026 - Present

- Sole software engineer owning the entire software function, including architecture, delivery, code review, and technical hiring support.
- Directing intern engineers and integrating across mechanical and electrical teams to build a modular ROS2 (Robot Operating System 2) codebase.
- Building production software at a robotics startup whose design-automation platform explores hardware and control-policy combinations using simulation and reinforcement learning.
- Developing and integrating ROS2 nodes for a fully autonomous construction tile-grouting robot deployed commercially in Singapore and Australia.
- Delivered a complete LiDAR sensor upgrade from integration through on-site deployment in one month.
- Integrating, tuning, and debugging across the autonomy stack including SLAM (Simultaneous Localisation and Mapping), motion planning, and Gazebo simulation.
- Containerizing robotics workloads with Docker for reproducible builds across development machines and on-robot deployment targets.
- Automating build, test, and release pipelines with GitHub Actions and CI/CD workflows.

MODEC Offshore Production Systems (Singapore) / Offshore Frontier Solutions
DevOps Intern | Singapore | Dec 2025 - Jan 2026

- Managed end-to-end deployment of an analytical data application from proof of concept to secure production.
- Configured and operated AWS Elastic Beanstalk environments with security-aligned release processes.
- Integrated CI/CD workflow and validation gates to maintain stable, repeatable delivery.
- Executed testing and rollback planning to satisfy organizational security and reliability benchmarks.

Lumerva
Co-Founder and Full Stack Engineer | Singapore | Sep 2025 - Present

- Building a production learning platform with full-stack ownership across React and TypeScript frontend and Python backend services.
- Designed asynchronous background-job architecture with Celery and Redis to improve throughput and reduce request latency.
- Implemented modular API and content pipelines for personalized course paths, quizzes, and content modules.
- Integrated external APIs including YouTube Data API and Brave Search API with ranking logic to improve discovery quality.

Misumi Kaizen OU Group (MISUMI.AI)
DevOps Engineer and Team Lead | Singapore | Sep 2025 - Aug 2026

- Led development and deployment of LISA 1.0, an automotive advisor platform built with Next.js 15, PostgreSQL, and AI-enabled retrieval workflows, deployed to production and used by MISUMI to serve customers internationally.
- Owned cloud architecture and infrastructure operations for production delivery across 4 ASEAN markets.
- Co-architected containerized microservices on Alibaba Cloud with a RAG pipeline over structured Japanese and English automotive listings.
- Reduced API latency from 300ms to 50ms through caching, query optimization, and service-level tuning.
- Drove performance optimization and workload tuning that reduced cloud costs by 70 percent.
- Transitioned to developer and technical advisor for LISA 2.0, supporting a new team on architecture and retrieval design.

National Institute of Education, Singapore
AI Engineer, Part-Time | Singapore | Jun 2025 - Oct 2025

- Developed backend features and real-time infrastructure for an intelligent tutoring system.
- Architected WebSocket-based low-latency communication patterns, achieving 99.9 percent session stability.
- Supported end-to-end testing and system hardening for robust production behavior.
- Organized user-study processes and consolidated qualitative and quantitative insights for product iteration.
- Co-authored a peer-reviewed Springer publication at AIED 2026 on scaffolding emotion regulation in productive failure, based on a 60-participant human-subjects study.

MODEC Offshore Production Systems (Singapore) / Offshore Frontier Solutions
Software Engineer Intern, Sustainability Analytics | Singapore | Jun 2025 - Aug 2025

- Independently delivered a full-stack web application for sustainability and environmental analytics.
- Built responsive frontend workflows in React for complex vendor data collection and reporting.
- Engineered a scalable backend in Django and Django REST Framework handling 500 or more concurrent requests with 99.9 percent uptime.
- Improved retrieval performance by 40 percent using Redis caching and optimized backend query paths.
- Managed AWS deployment and ingestion workflows integrating IoT logs via AWS Lambda and Amazon S3.

Panasonic R&D Center Singapore
Full Stack Engineer Intern | Singapore | Jan 2025 - Jun 2025

- Deployed an inference API using FastAPI and Docker serving YOLOv8 with async processing for real-time workflows.
- Debugged and optimized complex workflows across SQL, Python, Redis, and network-integrated services.
- Implemented and refined caching strategies to improve data retrieval speed, latency, and reliability.
- Designed and executed testing strategies to identify bottlenecks and validate system integrity.
- Monitored system performance and proposed data-driven enhancements to uptime and user experience.

NTU College of Computing and Data Science (CCDS)
Full Stack Developer | Singapore | May 2024 - Jul 2025

- Led end-to-end development of a networking platform connecting alumni and undergraduates for startup and event collaboration.
- Built and maintained full-stack features using WordPress, SQL, PHP, and frontend customization workflows.
- Optimized backend processes and site performance to improve reliability and user experience.

Schaeffler Hub for Advanced Research
Embedded Engineering Intern, Edge Data Systems | Singapore | May 2024 - Aug 2024

- Integrated the CAN bus protocol on an XMC4800 microcontroller, establishing a master-slave configuration between controller and actuator.
- Optimized performance by 20 percent leveraging low-latency EtherCAT and FreeRTOS scheduling on the XMC4800 microcontroller.
- Contributed to embedded framework development across Arduino, STM32F4, and XMC4800 environments.
- Performed debugging and validation to improve reliability of multi-board integration workflows.
- Enabled higher-frequency telemetry pipelines for predictive maintenance and edge analytics use cases.

Nanyang Technological University Singapore
SCSE Peer Coach | Singapore | Jan 2024 - Apr 2024

- Selected as one of four SCSE peer coaches to support students on academic warning.
- Delivered personalized study planning, mentorship, and accountability support to improve academic outcomes.

Freelance
Chess CCA Coach, MOE-qualified | Singapore | Jan 2024 - Apr 2024

- Delivered structured chess lessons during school CCA sessions, focusing on critical thinking and strategy.
- Built student-focused training plans to improve problem-solving and decision-making under constraints.

PROJECTS

MHC-Peptide Binding Prediction, Final Year Project
Bioinformatics Lab, NTU | Aug 2025 - May 2026

- Outperformed competing state-of-the-art models retrained on the same dataset and split, achieving a Pearson correlation coefficient (PCC) of 0.859 against a best baseline of 0.847.
- Improved R-squared and Spearman rank correlation by 0.5 to 2 percent over the strongest baseline.
- Constructed the dataset from IEDB (Immune Epitope Database), generating 3D structures with a self-developed algorithm extending ColabFold.
- Applied homology-based 80/10/10 train, validation, and test splits to prevent sequence-similarity data leakage.
- Integrated ESM-2 sequence embeddings with 650M parameters into a fine-tuned AlphaFold-based pipeline for binding affinity prediction.
- Supervisor: Professor Jagath C. Rajapakse.
- Orchestrated memory-intensive training runs on an NVIDIA A100 80GB GPU.
- Implemented a reproducible PyTorch training workflow with LoRA and PEFT, metric logging, and experiment tracking.

MyCCDS Student Platform
Production iOS and Android app for NTU CCDS students | Nov 2024 - Present

- Shipped a cross-platform mobile app using React Native, Expo, and Redux.
- Reached 600 or more daily active users with a 5-star store rating.
- Optimized caching to improve content discovery workflows.

Mango Order Management System
B2B ordering, payments, and operational analytics for a retailer | Jan 2025 - May 2025

- Built end-to-end ordering and payment workflows using React, Firebase, and Stripe.
- Processed more than SGD 18,000 monthly in transactions.

President Research Scholar, URECA
Edge AI security pipeline on ODROID | Aug 2024 - Apr 2025

- Achieved 96 percent motion detection accuracy with 180ms latency in an IoT security system on ODROID.
- Tuned motion pipeline thresholds and buffering to reduce false positives under changing lighting conditions.

SyncTube
Real-time synchronized video streaming and collaboration platform | Sep 2024 - Dec 2024

- Built synchronized playback and room-state management for multi-user watch sessions.
- Implemented real-time communication workflows and reconnect handling for stable room continuity.

Sentiment Analysis Platform
SC4001 Group Project, end-to-end text analytics with model-serving APIs | Jan 2024 - Apr 2024

- Built preprocessing and model-inference endpoints for sentiment classification workflows.
- Delivered dashboard views for trend analysis and data-driven business insights.

HACKATHONS AND COMPETITIONS

nGoDonate, EasyA x Ripple Hackathon
Winner, Best Developer Tooling | Jun 2025 - Nov 2025

- Built a donation workflow on XRPL and Ethereum achieving 2 second transaction finality.
- Implemented milestone-based escrow with on-chain transparency and NFT receipts and certificates.
- Added IPFS storage for verifiable supporting documents linked to on-chain records.

FunctionGemma, Cactus x Google Global Hackathon
National Top 10 | Feb 2025

- Engineered an agentic workflow using Google Gemma with function calling.
- Architected a modular Python API backend for rapid feature iteration.

Truth Fairy, TechFest Hackathon
Top 5 | Jan 2025

- Built a Chrome extension integrating an AI deepfake detection service with Grad-CAM explainability.
- Implemented lightweight inference orchestration and caching to keep analysis responsive.

GroceryGrabber, DSTA BrainHack
Finalist | May 2024

- Led a team and implemented PostGIS geospatial filtering within a 5km radius with GPT-powered parsing.
- Designed ranking logic for nearby inventory discovery and reduced redundant queries via server-side caching.

Port Downtime and Freight Forecasting, PSA Code Sprint
Finalist | May 2024

- Built a TensorFlow time-series forecasting pipeline with a Django backend and operational dashboard.

CERTIFICATIONS

- Meta Front-End Developer Specialization
- HTML and CSS in Depth
- Introduction to Front-End Development
- Introduction to Generative AI
- Version Control

AWARDS AND HONORS

- Lee Kuan Yew Gold Medal, Nanyang Technological University
- Honours (Highest Distinction), highest degree classification awarded by NTU
- Peer-reviewed publication, AIED 2026, Springer Nature Switzerland
- NTU Science and Engineering Undergraduate Scholarship, Full Scholarship
- Dean’s List, all four academic years
- President Research Scholar, URECA
- Winner, Best Developer Tooling, EasyA x Ripple Hackathon
- National Top 10, Cactus x Google Global Hackathon
- Finalist, DSTA BrainHack
- Finalist, PSA Code Sprint
- Malaysia National Cubing Open Finalist

- ABRSM Piano Grade 8, Distinction

LEADERSHIP AND ACTIVITIES

- President and Captain, NTU International Chess Varsity Team
- Music Director, NTU Piano Ensemble
- NTU SCSE Peer Coach

LANGUAGES

- English - Full Professional
- Tamil - Native or Bilingual
- Hindi - Professional Working
- Spanish - Limited Working
- Sanskrit - Elementary

ADDITIONAL KEYWORDS

- Full-Stack Architecture, Cloud Infrastructure, DevOps Leadership, Site Reliability, Microservices, Distributed Systems
- Frontend Development, Backend Development, API Integration, Payment Integration, Airwallex, Stripe, Real-time Systems
- Machine Learning Engineering, MLOps, Large Language Models, Retrieval Augmented Generation, Agentic AI, Edge AI, Internet of Things
- Robotics Software Engineering, ROS, ROS2, Autonomous Systems, Reinforcement Learning, Simulation, Bioinformatics, Protein Language Models, Peer-Reviewed Publication
- Startup Leadership, Startup Ventures, Product Ownership, Cross-Functional Collaboration, Technical Mentorship, Cost Optimization